

(Zou, 2024), the latent activations across the validation set were normalised using MinMax scaling. In this instance, similar to Simon and Zhou, we took the mean activation of the latents across all residues in the sequence. For each normalised latent, binary latent-on/latent-off labels using activation thresholds of 0.1, 0.2, 0.5, 0.8, 0.9 were applied. For each latent–concept pair, we defined a latent as an interpretable feature only if it achieved an F_1 score that exceeded a class-prevalence baseline by a fixed margin. Let π denote the prevalence of the positive class for the concept in the validation set. We first computed the F_1 score of a trivial always-positive classifier,

$$F_{1,\text{all}+} = \frac{2\pi}{1 + \pi}. \quad (6)$$

A latent was labelled a feature if, for any tested activation threshold, its F_1 satisfied

$$F_1 > F_{1,\text{all}+} + 0.2 \quad (7)$$

This criterion accounts for class imbalance and requires each latent to provide predictive signal beyond a strong non-informative baseline, rather than exploiting the marginal class distribution. It is worth noting that the threshold for ‘feature’ is somewhat arbitrary. Latents just below the threshold are likely to still have meaningful correlations to a target concept.

3.2.4. ANTIBODY SEQUENCE ALIGNMENT

Antibody sequences were aligned using ANARCI (Dunbar & Deane, 2016) and the IMGT numbering (Lefranc et al., 2003).

3.3. Steering

Steering was implemented based on the strategy by Templeton et al. (Templeton et al., 2024). Each latent can be represented by its corresponding decoder vector $d(i) = W_{\text{dec}}[i, :]$, where $d(i)$ is the decoder vector for latent i and W_{dec} is the decoder weight matrix. Steering is performed by scaling the decoder vector and adding it to the original hidden state (Equation 8).

$$h_l^* \leftarrow h_l + \alpha \cdot d(i) \quad (8)$$

Here, α is the steering factor and h_l is the hidden state before the intervention and h_l^* is the hidden state following the intervention.

4. Results

4.1. TopK SAE-identified Features are Interpretable, Antibody-specific Concepts, but Not Steerable

4.1.1. TOPK SAE LATENTS PRESERVE BIOLOGICAL INFORMATION FOLLOWING SPARSIFICATION

TopK sparsification represents each token with far fewer latents compared to the hidden neurons, creating a possibility for information loss during sparsification. To qualitatively check whether antibody-specific information is retained within the latents, we compared the accuracies of using latents and hidden neurons on residue and sequence-level property prediction tasks.

We began the analysis by focusing on TopK SAEs trained on the final layer activations (layer 3), as seen in previous works (Parsan

et al., 2025a), since they contain the most complete representation of the sequence. Logistic regressor training on CDR identities using latent activations obtained a validation accuracy of **0.99**, while using hidden neuron activations resulted in a validation accuracy of **0.98**. This showed that residue-level CDR identity information is preserved in the activated latents.

In order to check sequence-level features, we investigated germline gene predictions. We focused on heavy J genes for simplicity, given they have less allelic variation compared to heavy V genes. LR training resulted in a validation F_1 macro score of **0.93**. We used an F_1 score due to significant class imbalance in heavy J genes within the training and validation sets. Table 1 reports the precision, recall and F_1 -scores of the different gene identities studied and shows how the model accurately predicts each IGHJ class using the SAE latents. Overall, the high F_1 macro score shows how the activated latents also preserve sequence-level features.

Our results indicate that SAE latents collectively represent antibody information following sparsification when applied to an antibody language model, similar to results seen in previous works for general protein language models (Simon & Zou, 2024; Adams et al., 2025; Parsan et al., 2025a). We therefore carried out further analysis of SAE latents to investigate their overall interpretability.

Table 1. Precision, recall, and F_1 -score per IGHJ class.

Class	Precision	Recall	F_1 -score
IGHJ1	0.91	0.71	0.80
IGHJ2	0.94	0.93	0.94
IGHJ3	0.96	0.96	0.96
IGHJ4	0.98	0.99	0.98
IGHJ5	0.94	0.95	0.95
IGHJ6	0.98	0.97	0.98
Macro average	0.95	0.92	0.93

4.1.2. TOPK LATENT ACTIVATIONS ARE VISUALLY INTERPRETABLE

To investigate whether SAE latents provide a tangible method of interpreting the logic being model generation, we compared the activated patterns of latents and neurons correlated to properties of interest. As a baseline, we compared activations of the top correlated latents and neurons for CDRH3.

Visual investigation showed that latent activations are sparse and specific to CDRH3 residues, compared to neurons which activate across the sequence without any immediately recognisable pattern (Figure 1). This may be explained by the polysemanticity of neurons, where multiple features specific to several unrelated residues are represented by the same neuron. When investigating activation patterns, this complicates using neurons as a tool for interpretability and highlights the potential greater explainability of SAE-derived latents.

In order to investigate the utility of SAE activations for the mechanistic interpretability of sequence-level concepts, as opposed to residue-level concepts, we investigated heavy J gene activations

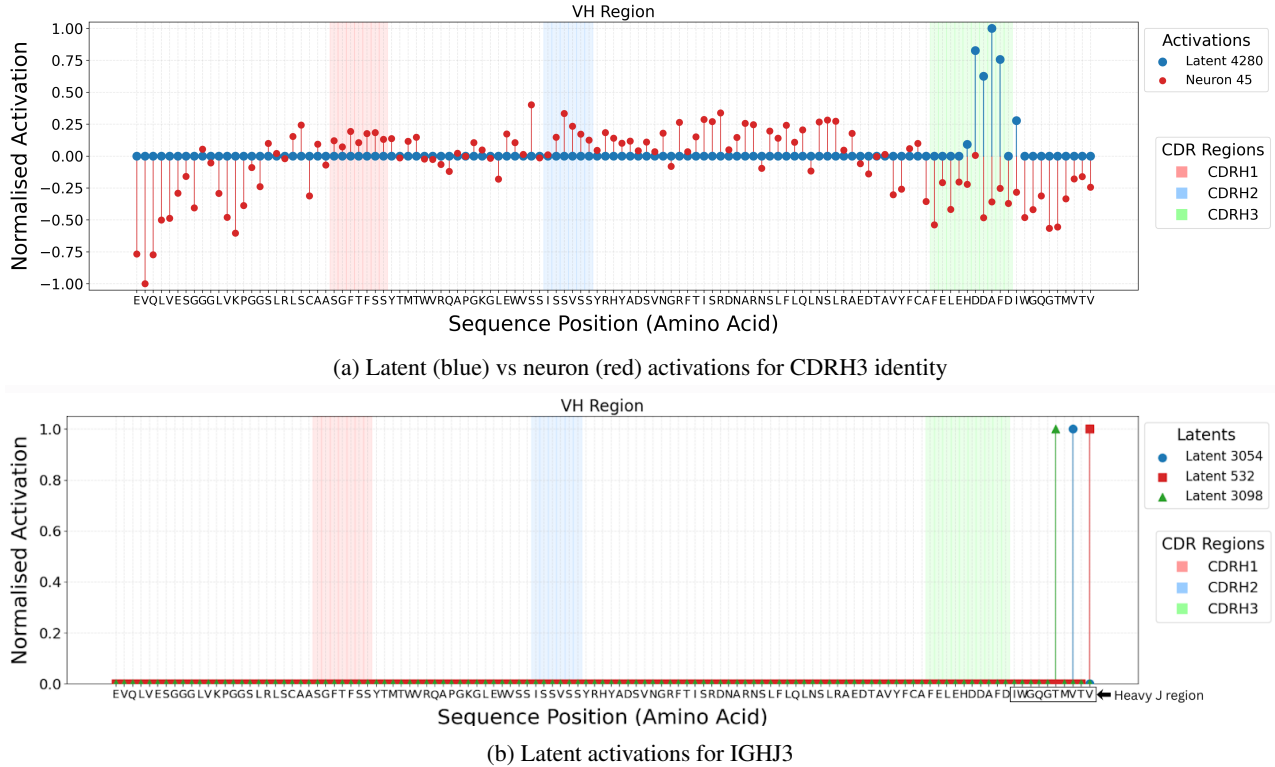


Figure 1. Latent activations (blue) and hidden neuron activations (red) for CDRH3 (a), and latent activations for IGHJ3 (b). The x-axis shows the amino-acid sequence of the VH region of a test antibody; the y-axis shows normalised activation. CDRs are coloured CDRH1 (red), CDRH2 (blue), and CDRH3 (green). Latent activations localise to the expected regions—CDRH3 in and the heavy J region, whereas neuron activations are scattered across the sequence with no discernible pattern.

as sequence-level concepts. Similar to residue-level observations, latents corresponding to heavy J gene identity activate on residues representing the concept, i.e. gene identity. In this instance, the top correlated latents were activated on the J domain of examined antibody sequences. Sequence-level representations are mean pools of the original residue-level representations, leading to an overall loss of positional information. Therefore, the top correlated latents also encode intrinsic positional information (Figure 1c). This provides an opportunity to identify the residues responsible for a global, sequence-level feature, with potential implications for the understanding of the sequence and structural basis of antibody properties.

To quantify the predictive properties of our identified features, we carried out an activation-threshold analysis (Supplementary Table 4, Methods 3.2.3).

4.2. Ordered SAEs Identify More Steerable Features Compared to TopK SAEs

4.2.1. P-IGGEN GENERATION CAN BE PREDICTABLY STEERED USING ORDERED SAE LATENTS

In addition to assessing how well latents predict the target concept, we also tested feature steering as an indicator of feature importance (Parsan et al., 2025a): if increasing a latent consistently steers generation in a desired direction, the corresponding

feature is likely important for that direction. We were unable to successfully steer on TopK latents within p-IgGen’s latent space (See Supplementary A.1), which may be attributed to known issues within these architectures such as feature splitting and absorption (Chanin et al., 2024). We subsequently evaluated Ordered SAEs, which build a hierarchical latent space that preserves both high-level and fine-grained features (Bussmann et al., 2025), but result in a less interpretable localisation pattern (Supplementary Figure 7).

We conducted a linear probe and subsequent activation-threshold analysis to identify features correlated to IGHJ4 in layer 3. Due to the implicit hierarchy in features, we ranked latents based on their dictionary index, with smaller indices representing higher-level features. We identified latent 12, which was positively associated with IGHJ4 based on activation-threshold analysis, and latent 49, which had a strongly negative logistic regression coefficient (Figure 2).

Positively steering on latent 12 increased IGHJ4 proportion in model generation (Pearson’s $R = 0.939$, $p = 6.458 \times 10^{-7}$ and Spearman’s correlation = 0.921, $p = 2.982 \times 10^{-6}$). Conversely, positively steering on latent 49 decreased IGHJ4 proportion in model generation (Pearson’s R of -0.705, p -value = 4.89×10^{-3} and Spearman’s correlation = -0.657, $p = 0.0106$).

To investigate the features, we plotted their activations on spe-

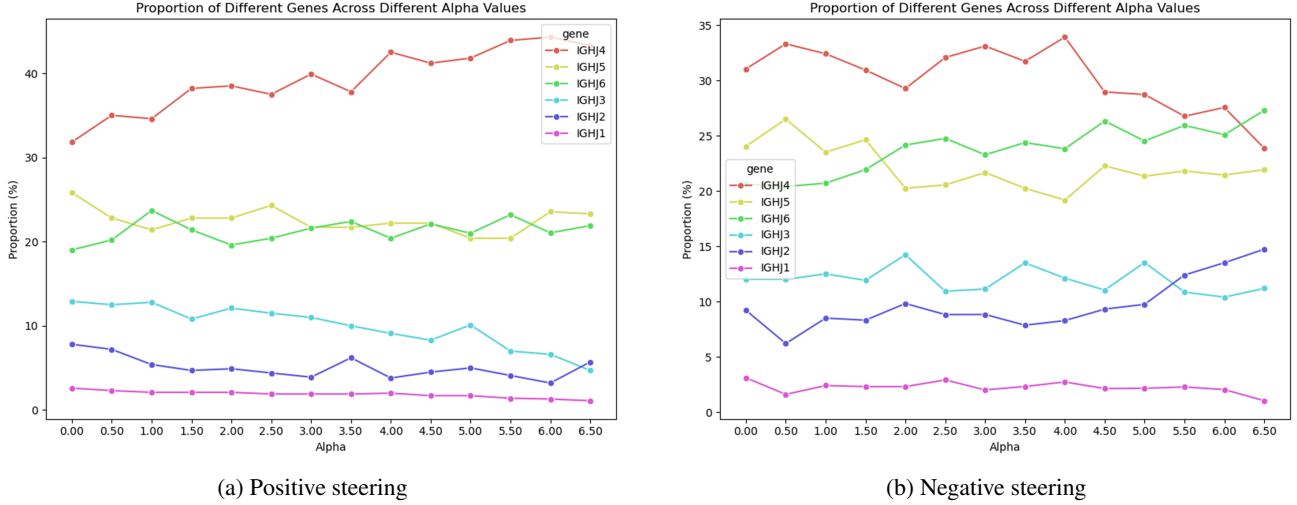


Figure 2. Results of IGHJ4 steering using O-SAE latent 12 (a) and 49 (b). Y-axis shows the proportion of generated sequences. Plots are coloured by heavy J gene identity. X-axis shows the steering factor used (alpha). Results are for a library of 1000 p-IgGen-generated sequences. Latent 12—positively associated with IGHJ4 based on activation-threshold analysis—increases IGHJ4 proportion under positive steering, whereas latent 49—which has a strongly negative logistic regression coefficient—decreases IGHJ4 under the same steering.

cific IMGT positions across a random set of IGHJ4 containing sequences (Supplementary Figure 7) The latents activate more broadly across the sequence than TopK latents. One reason for this may be the abstraction which enables steerability. The latents inform downstream generation, and therefore signal early on during sequence generation about the identity of the J-region. Given the latents communicate longer-range concepts, it is possible that latent activation is not linked to any specific residue, rather a range of residues within a defined length of the sequence.

4.2.2. NESTED STEERING GENERATES NOVEL, DIVERSE, ANTIBODY-LIKE SEQUENCES

To assess whether nested steering alters the generative distribution while retaining antibody-like properties, we evaluated steered libraries using sequence novelty (Hamming distance to the held-out validation set), diversity (nearest-neighbour cosine distance), and protein-likeness (ESM-2 likelihood). Across all metrics, steered sequences remained close to the natural baseline, with only modest absolute shifts (Table 2).

Steered sequences using O-SAE latent 12 (Figure 2 (a) and alpha 6.50) were slightly more distant from the validation set than natural sequences for both VH (24.67 vs. 22.91; $\Delta \approx 1.76$) and VL (20.24 vs. 19.81; $\Delta \approx 0.43$), indicating increased novelty. Steering also produced a small increase in diversity (0.23 vs. 0.20; $\Delta \approx 0.03$) Protein-likeness was essentially unchanged, with ESM-2 likelihoods differing by only ~ 0.005 (-0.325 vs. -0.330), suggesting steered sequences remain on a highly plausible antibody manifold. ANARCI (Dunbar & Deane, 2016) could number all 1000 of the generated sequences and distinguish between the heavy and light chains, further indicating antibody-likeness.

For statistical testing, we compared the distribution of each metric between steered and natural sequences using a two-sided Mann–

Table 2. Steered generation analysis. Asterisks indicate a statistically significant difference between steered and natural sequences under a two-sided Mann–Whitney U test ($p < 0.05$).

Metric	Steered	Natural	Δ
Mean VH Hamming dist. to val. set	24.67*	22.91	1.76
Mean VL Hamming dist. to val. set	20.24*	19.81	0.43
Diversity	0.23*	0.20	0.03
ESM2-likelihoods	-0.325^*	-0.330^*	0.005

Whitney U test. Although all comparisons were statistically significant ($P < 0.05$), effect sizes were uniformly small: rank-biserial correlations were < 0.2 for each metric, indicating substantial overlap between the steered and natural distributions.

4.3. SAE-Based Interpretability Transfers Across Antibody Language Models

In order to demonstrate the generaliseability of SAEs for mechanistic interpretability studies, we examined two additional antibody language models of different scale and architecture: (i) **IgLM** (4 layers, 512 hidden dimensions, 13M parameters) and (ii) **ProGen2-OAS** (27 layers, 1536 hidden dimensions, 764M parameters).

Using the TopK pipeline on final-layer activations, we identified SAE latents that are maximally predictive of IGHJ germline identity. High-scoring features were found for IGHJ1 (ProGen2-OAS: max $F_1 = 0.535$), IGHJ3 (IgLM: max $F_1 = 0.641$; ProGen2-OAS: max $F_1 = 0.775$), IGHJ4 (IgLM: max $F_1 = 0.692$; ProGen2-OAS: max $F_1 = 0.679$), and IGHJ6 (IgLM: max $F_1 = 0.773$; ProGen2-OAS: max $F_1 = 0.692$). In contrast, no strong TopK features were detected for IGHJ2 or IGHJ5.